



COMPLETE

NODE JS

FUNDAMENTALS

- ***Event-Driven Architecture***: Uses events and callbacks for asynchronous operations.
- ***Non-Blocking I/O***: Doesn't wait for operations to complete before moving to the next task.
- ***Global Objects***: process, console, Buffer, `__dirname`.
- ***Modules***: require (CommonJS) vs import (ES Modules).

ASYNCHRONOUS PROGRAMMING

- **Callbacks:** *Functions passed as arguments to handle async results.*
- **Promises:** *Handle async operations with `.then()` and `.catch()`.*
- **Async/Await:** *Syntactic sugar over Promises for cleaner code.*
- **Event Loop:** *Mechanism that handles async callbacks in phases (timers, I/O, etc.).*

FILE SYSTEM & STREAMS

- ***fs Module: Read/write files synchronously or asynchronously.***
- ***Streams: Handle large data chunks efficiently (Readable, Writable, Pipe).***

NETWORKING

- ***HTTP/HTTPS Modules: Create servers and make requests.***
- ***RESTful APIs: Design endpoints for CRUD operations.***
- ***WebSockets: Real-time communication (socket.io or ws).***

DEBUGGING & PERFORMANCE

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- *--inspect Flag: Debug with Chrome DevTools.*
- *Clustering: Utilize multi-core CPUs with the cluster module.*
- *Memory Leaks: Identify using heap snapshots.*

SECURITY

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- ***Helmet.js: Secure HTTP headers.***
- ***Input Validation: Sanitize user inputs to prevent attacks.***
- ***Rate Limiting: Prevent brute-force attacks.***

DATABASES

- ***MongoDB: NoSQL database with Mongoose ODM.***
- ***SQL: PostgreSQL/MySQL with Sequelize or TypeORM.***
- ***Redis: Caching and real-time features.***

TESTING

- ***Unit Tests: Jest/Mocha for isolated function testing.***
- ***Integration Tests: Supertest for API endpoints.***

INTERVIEW QUESTIONS

BEGINNER-LEVEL QUESTIONS

- 1. What is Node.js, and how does it differ from traditional server-side platforms?**
- 2. Explain the event loop in Node.js.**
- 3. How do you handle file operations in Node.js?**
- 4. What is the purpose of package.json?**

INTERMEDIATE LEVEL QUESTIONS

- 1. How do Promises improve callback hell?**
- 2. Explain the difference between `setImmediate` and `process.nextTick`.**
- 3. How would you create a simple HTTP server?**
- 4. What is middleware in Express.js?**

ADVANCE LEVEL QUESTIONS

- 1. How does Node.js handle child threads?**
- 2. Explain clustering in Node.js.**
- 3. How would you debug a memory leak?**
- 4. Implement JWT authentication in Express.**